

Project Title: SLAIT – Secure Language Assembly Inspector Tool

Team Members:

- Maria Linkins-Nielsen – mlinkinsniel2022@my.fit.edu
- Michael Bratcher – mbratcher2021@my.fit.edu

Faculty Advisor: Dr. Marius Silaghi – msilaghi@fit.edu

Client: Dr. Marius Silaghi – CSE Professor & Faculty Advisor

Progress of Current Milestone:

Task	Completion%	Michael	Maria	Reasoning/To-Do
Deploy Build to Production Server	100%	50%	50%	Build is operational on the production server.
User Testing	75%	50%	50%	Peer user testing and review has occurred but testing within the targeted class has not due to delays in production server deployment
Final Touches (Templates, User Testing Bug Fixes)	100%	25%	75%	Multiple last minute requests and additions were implemented successfully
Finalizing System Documentation & Demo	75%	50%	50%	Documentation is complete, but system demo video is still required

Discussion of Milestone Tasks:

1.

The current build has been deployed onto the fac-silaghi-m1 server, and is accessible <http://172.16.135.14/> while on the eduroam subnet via wifi or connect via ethernet. Basic function testing occurred and ensured expected functionality occurs as it does on the development server. The development server has been phased out. Permissions for the application are tied to the mbratcher2021 user permissions on the server, so if for whatever reason those permissions are rolled back or unallocated, the system may not deliver to the backend, as the systemctl process “slait” runs actions with admin permissions through the user account, not root itself.

2.

We’ve performed and received peer review of our project, but our main test goal is the Assembly/Machine Language class to perform a larger more practical user test on testers who are actively working with the language.

3.

Multiple final additions were requested and applied. This includes the following:

Snippet Button

This button is a tertiary option for the Input/Document widget, where the user may insert parts of the code into .data and .text boxes respectively, and upon pressing “done”, the file is created and combines the two “snippets” into one file, allowing the user to place lines onto a premade template without modifying an existing one.

Inspection Modification

Users now have two extra functions to interact with once they create an inspection: “Duplicate”, and “Edit”. “Duplicate” creates a new inspection with the duplicated inspection’s information maintained within the new inspection, only requiring the user to provide a new line number instead of filling in every flag again. “Edit” takes the inspection you press “Edit” on and places it in the pre-done state again, maintaining its information. This allows users to fix mistakes or add extra variables without having to delete and make an entirely new inspection.

Widget Resizing and Modifying Output Widget Content Order

The input and inspection widgets are no longer variably tied to each other in terms of sizing, and are now a static size. The table output visual has been placed at the top of the widget per request.

Formalizations

Florida Tech logos were added to the page header and tab icon.

4.

The senior design showcase poster has been completed and submitted, as well as the virtual showcase information. Tasks left to complete are combining our respective frontend/backend GitHub repositories into a unified repo for ease of deployment in future releases, as well as completing user guides, which we don't anticipate taking long, since the workflow is simple.

Contribution to Milestone Tasks:

Michael:

I imported and set up the project onto FIT servers, which included building and serving the frontend, building the backend image, and setting up the API system process to allow the system to deliver information to the backend without needing a user session running the script. I also performed the following final touch changes: Snippet Button, Inspection Modification, Widget Resizing and Modifying Output Widget Content Order, and Formalizations.

Maria:

I created the Senior Design Showcase Poster and Virtual Showcase Form. I also updated our system diagram to reflect current deployment setup.

1. Lessons Learned Over Course of Design

- 1. Importance of realistic goal setting
 - i. Proper goal setting is an important aspect of working on a complex project. Within the milestones of this project, we had some milestones where we set extremely high expectations, not realizing how much this can bottleneck important small provisions like documentation, as you either work to accomplish an excessive amount of subtasks required to accomplish the goal, or simply overwhelming your team with tasks you may end up rolling back on if the vision changes for any reason. At the same time, setting goals too small simply means your project will come to a crawl, unnecessary additions may conflict with important requirements, and time to complete may take priority over importance of completion, leaving major functionality come down to wire. Finding a good balance is important, and the next lesson learned can help with that.
- 2. Using Project Management Applications (JIRA Framework)
 - i. We utilized the JIRA by Atlassian framework to keep track of progress and items that needed to be addressed in their different stages. This helped significantly early on with the first milestone documentation, as keeping track of the status of every item becomes easier with a system

designed for it. JIRA helped us manage the stages of our project, and make work apparent even during design bottlenecks

- 3. Hosting and Serving Webpages
 - i. Neither of us have served a webpage before, and this project was a first. We learned the basic of Apache2, local host serving vs local network serving, and transporting data between different locations in a server using flask scripts.
 - 4. GDB
 - i. Most of our experience surrounding GDB has been for use in binary analysis or reverse engineering purposes. Combining use of GDB with Python scripting proved to be a challenge, but a good learning opportunity.
 - ii. It was unknown at the time of project development, but GDB has a built in tool for Python scripting as well as command files that work similarly. We are not sure if these would have saved time for our project development, but it would have been an interesting tool to use.
 - 5. Frontend Web Development
 - i. Besides HTML and some JavaScript, this was my first project making a fully functional frontend design. I learned the fundamentals of Apache and its widget-based design to successfully build multiple complex components.
2. Date(s) of meeting(s) with Client during the current milestone: 04/08/2026
3. Client feedback on the current milestone
- see Faculty Advisor Feedback below
4. Date(s) of meeting(s) with Faculty Advisor during the current milestone:
5. Faculty Advisor feedback on each task for the current Milestone
- Task 1:
 - Task 2:
 - Task 3:

- Task 4:

6. Faculty Advisor Signature: _____ Date: _____

7. Evaluation by Faculty Advisor

- Faculty Advisor: detach and return this page to Dr. Chan (HC 209) or email the scores to pkc@cs.fit.edu
- Score (0-10) for each member: circle a score (or circle two adjacent scores for .25 or write down a real number between 0 and 10)

Michael	0	1	2	3	4	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10
Maria	0	1	2	3	4	5	5.5	6	6.5	7	7.5	8	8.5	9	9.5	10

○ Faculty Advisor Signature: _____ Date: _____